

①

Traped Water

			Drop	1	← Height
1	○	○	○	2	
2	○	1	○	3	
3	○	2	○	4	
a[5] =	3	0	2	0	4

Time complexity = $O(n^2)$ Space complexity = $O(1)$

a[5] =

minimum (leftman, rightman)
minimum - height

// Pseudo code

```
for (i = 0; i < 5; i++)
{
```

```
    leftman = a[0];
```

```
    for (j = 0; j <= i; j++)
    {
```

```
        if (leftman < a[j])
```

```
        {
            leftman = a[j];
        }
    }

```

```
    rightman = a[i];
```

```
    for (j = i; j < 5; j++)
    {
```

```
        if (rightman < a[j])
```

```
        rightman = a[j];
    }

```

```
    minimum = (leftman < rightman) ? leftman : rightman;
    water += minimum - a[i];
}
```


Trapped water by Two pointer approach

(1) left = 0 (index)

$$\text{left max} = 0 \text{ (1m)}$$

// Pseudo Code

(2) right, 4

right max: $O(n)$

water = 0 (w)

while ($l < r$)

5

if ($a[l] < a[r]$)

$$\{ iF[a[l]] \geq lm \}$$
$$lm = a[l];$$

eise

$$wt = lm - a[l];$$

l++;

3

else

3

if $(a[x] > = y_m)$

3

$$Am = a[n];$$

3

else

$$\omega + = \gamma m - q[\phi];$$

2--;

3

3

Then print w.

Time Complexity $\Rightarrow O(n)$

Space Complexity $\Rightarrow O(1)$